ytdu@purdue.edu
+1 765 543 3011
www.yuntaodu.me
Google Scholar

Yuntao Du

Research Interests

AI Security & Privacy; Data Privacy; Trustworthy Machine Learning.

Education

Purdue University 2023 – 2027 (expected)
Ph.D. in Computer Science (Advisor: Prof. Ninghui Li) USA

Zhejiang University 2020 – 2023
M.E. in Computer Technology. *Rank 1/49.* China

East China Normal University 2015 – 2020
B.E. in Data Science. *GPA 3.61/4 (rank 1/23).* China

Selected Honors and Awards

National Winner, Innovation Bowl (Purdue Team Lead), covered by Purdue CS News 2026
NDSS Fellowship, Internet Society (1 of 24 worldwide) 2026
University Nominee, Google PhD Fellowship 2026
University Nominee, Apple Scholars in AI/ML PhD Fellowship 2026
Ross Fellowship, Purdue University 2023
Herbold Scholarship, Purdue University (1 of 7) 2023
Presidential Doctoral Excellence Award, Purdue University (1 of 150) 2023
Excellent Master's Dissertation, China (1 of 43) 2023
Provincial Outstanding Graduate, Zhejiang, China 2023
National Scholarship, China (0.1%) 2021, 2022

Professional Experience

Research Intern, Perplexity AI 2026.05 – Present
Mentor: Dr. Ninghui Li San Francisco, USA

Investigated privacy and security risks of real-world AI agents and co-authored a technical report on agent sandbox security; developed a risk-based access control framework that informed the design of secure AI agent systems in Perplexity's products.

Research Intern, Alibaba Qwen 2024.05 – 08
Mentor: Dr. Bolin Ding Seattle, USA

Developed an agentic framework to investigate emerging privacy risks from AI misuse, resulting in a paper at ACL Findings 2026 and a pioneering survey of LLM privacy risks published in the IEEE Data Engineering Bulletin.

Research Intern, Microsoft Research Asia 2022 – 2023
Mentor: Dr. Xing Xie Beijing, China

Developed explainable collaborative filtering deployed in the Xbox game recommendation system, resulting in a WWW 2023 paper and a U.S. patent; received the Excellent Internship Award (top 10%).

Research Proposal Development

NSF SaTC 2.0 Submitted, 2026	**Reference-Grounded AI Data Accountability** Co-wrote Thrust 1 on data-use detection with Prof. Hanshen Xiao and Prof. Ninghui Li, based on my membership inference research [P16], [P20].
NSF SaTC 2.0 Submitted, 2026	**Model Robustness, Ownership, and Privacy-preserving in Transfer Learning** Co-wrote the privacy-preserving transfer learning section with Prof. Elisa Bertino and Prof. Ninghui Li, drawing on my research on data synthesis [P15] and privacy auditing on LLMs [P13].
NSF ACTION Mini-proposal, 2026	**Exploring Tokenizer-based Attack Surfaces for LLMs** Authored the full mini-proposal building on my tokenizer privacy research [P19], in collaboration with Prof. Ninghui Li and Prof. Sewoong Oh (University of Washington).
Cisco Research Award, 2025 Funded; $60,000	**Quantifying and Characterizing Privacy Leakage in Vision-Language-Action Models via Membership Inference** Co-wrote the proposal with my advisor, Prof. Ninghui Li, developing the research plan for extending my privacy auditing methods [P16], [P18] to vision-language-action models.

Publications

First/Co-First	**11** publications at CCS (×2), USENIX Security (×2), NDSS, VLDB, ACL Findings, WWW, SIGIR, TKDE, and IEEE Data Engineering Bulletin.
Mentoring	Students I mentored have **10** papers at CCS (×2), USENIX Security, VLDB, SIGIR (×3), ICDE (×2), and TKDE.

*equal contribution, †student mentored

[P22] CCS 2026 — **Privacy Leakage from a Thousand Words: Millipixel Location Recovery from Dot Maps.**
Yuntao Du[*], Tanishq Pauskar[†*], Hao Wang, Jing Su, Ninghui Li.
In ACM Conference on Computer and Communications Security (CCS), 2026.

[P21] ACL 2026 Findings — **Automated Profile Inference with Language Model Agents.**
Yuntao Du, Zitao Li, Bolin Ding, Yaliang Li, Hanshen Xiao, Jingren Zhou, Ninghui Li.
In Findings of the Association for Computational Linguistics (ACL), 2026.

[P20] USENIX 2026 — **Imitative Membership Inference Attack.**
Yuntao Du, Yuetian Chen, Hanshen Xiao, Bruno Ribeiro, Ninghui Li.
In USENIX Security Symposium (USENIX Security), 2026.

[P19] USENIX 2026 — **Membership Inference Attacks on Tokenizers of Large Language Models.**
Meng Tong[†*], **Yuntao Du**[*], Kejiang Chen, Weiming Zhang, Ninghui Li.
In USENIX Security Symposium (USENIX Security), 2026.

[P18] USENIX 2026 — **Window-based Membership Inference Attacks Against Fine-tuned Large Language Models.**
Yuetian Chen, **Yuntao Du**, Kaiyuan Zhang, Ashish Kundu, Charles Fleming, Bruno Ribeiro, Ninghui Li.
In USENIX Security Symposium (USENIX Security), 2026.

[P17] ICLR 2026 — **Membership Inference Attacks Against Fine-tuned Diffusion Language Models.**
Yuetian Chen, Kaiyuan Zhang, **Yuntao Du**, Edoardo Stoppa, Charles Fleming, Ashish Kundu, Bruno Ribeiro, Ninghui Li.
In International Conference on Learning Representations (ICLR), 2026.

[P16] NDSS 2026 — **Cascading and Proxy Membership Inference Attacks.**
Yuntao Du, Jiacheng Li, Yuetian Chen, Kaiyuan Zhang, Zhizhen Yuan, Hanshen Xiao, Bruno Ribeiro, Ninghui Li.
In Network and Distributed System Security Symposium (NDSS), 2026.

[P15] CCS 2025 **Systematic Assessment of Tabular Data Synthesis.**
Yuntao Du, Ninghui Li.
In ACM Conference on Computer and Communications Security (CCS), 2025.

[P14] IEEE Data Engineering Bulletin **Beyond Data Privacy: New Privacy Risks for Large Language Models.**
Yuntao Du, Zitao Li, Ninghui Li, Bolin Ding.
In IEEE Computer Society Data Engineering Bulletin, 2025.

[P13] USENIX 2025 **SOFT: Selective Data Obfuscation for Protecting LLM Fine-tuning against Membership Inference Attacks.**
Kaiyuan Zhang, Siyuan Cheng, Hanxi Guo, Yuetian Chen, Zian Su, Shengwei An, **Yuntao Du**, Charles Fleming, Ashish Kundu, Xiangyu Zhang, Ninghui Li.
In USENIX Security Symposium (USENIX Security), 2025.

[P12] ICDE 2024 **Real-Time Trajectory Synthesis with Local Differential Privacy.**
Yujia Hu†, **Yuntao Du**, Zhikun Zhang, Ziquan Fang, Lu Chen, Kai Zheng, Yunjun Gao.
In IEEE International Conference on Data Engineering (ICDE), 2024.

[P11] VLDB 2023 **LDPTrace: Locally Differentially Private Trajectory Synthesis.**
Yuntao Du, Yujia Hu†, Zhikun Zhang, Ziquan Fang, Lu Chen, Baihua Zheng, Yunjun Gao.
In International Conference on Very Large Data Bases (VLDB), 2023.

[P10] WWW 2023 **Towards Explainable Collaborative Filtering with Taste Clusters Learning.**
Yuntao Du, Jianxun Lian, Jing Yao, Xiting Wang, Mingqi Wu, Lu Chen, Yunjun Gao, Xing Xie.
In ACM Web Conference (WWW), 2023.

[P9] SIGIR 2023 **Knowledge-refined Denoising Network for Robust Recommendation.**
Xinjun Zhu†, **Yuntao Du**, Yuren Mao, Lu Chen, Yujia Hu, Yunjun Gao.
In International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2023.

[P8] ICDE 2023 **FLBooster: A Unified and Efficient Platform for Federated Learning Acceleration.**
Zhihao Zeng†, **Yuntao Du**, Ziquan Fang, Lu Chen, Shiliang Pu, Guodong Chen, Hui Wang, Yunjun Gao.
In IEEE International Conference on Data Engineering (ICDE), 2023.

[P7] TKDE 2023 **Finding Materialized Models for Model Reuse.**
Minjun Zhao, Lu Chen, Keyu Yang, **Yuntao Du**, Yunjun Gao.
In IEEE Transactions on Knowledge and Data Engineering (TKDE), 2023.

[P6] TKDE 2023 **MetaKG: Meta-learning on Knowledge Graph for Cold-start Recommendation.**
Yuntao Du, Xinjun Zhu†, Lu Chen, Ziquan Fang, Yunjun Gao.
In IEEE Transactions on Knowledge and Data Engineering (TKDE), 2023.

[P5] KDD 2022 **Spatio-Temporal Trajectory Similarity Learning in Road Networks.**
Ziquan Fang, **Yuntao Du**, Xinjun Zhu, Danlei Hu, Lu Chen, Yunjun Gao, Christian S. Jensen.
In ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2022.

[P4] SIGIR 2022 **Self-Guided Learning to Denoise for Robust Recommendation.**
Yunjun Gao, **Yuntao Du**, Yujia Hu†, Lu Chen, Xinjun Zhu, Ziquan Fang, Baihua Zheng.
In International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2022.

[P3] SIGIR 2022 **HAKG: Hierarchy-Aware Knowledge Gated Network for Recommendation.**
Yuntao Du, Xinjun Zhu†, Lu Chen, Baihua Zheng, Yunjun Gao.
In International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR), 2022.

[P2] ICDE 2021 **E^{2}DTC: An End to End Deep Trajectory Clustering Framework via Self-Training.**
Ziquan Fang, **Yuntao Du**, Lu Chen, Yujia Hu, Yunjun Gao, Gang Chen.
In IEEE International Conference on Data Engineering (ICDE), 2021.

[P1] VLDB 2021 **MDTP: A Multi-source Deep Traffic Prediction Framework over Spatio-Temporal Trajectory Data.**
Ziquan Fang, Lu Pan, Lu Chen, **Yuntao Du**, Yunjun Gao.
In International Conference on Very Large Data Bases (VLDB), 2021.

Peer-Reviewed Short Papers

[S1] CCS 2026 **A Preliminary Study of LLM Distillation Inference.**
Edward Chen†, **Yuntao Du**.
In ACM Conference on Computer and Communications Security (CCS), poster paper, 2026.

Manuscripts and Preprints

[M4] Under review **RATION: Towards Quantitative Risk-Based Access Control for AI Agents.**
Yuntao Du, Harsh Chaudhari, Yuetian Chen, Ron Heichman, Ashish Jayamohan, Kumar Pallav, Mahim Patel, Kaiyuan Zhang, Kyle Polley, Jerry Ma, Ninghui Li.
Manuscript under review, arXiv:2610.24311, 2026.

[M3] Under review **Phantom Membership Inference Attack.**
Yuntao Du, Yuetian Chen, Hanshen Xiao, Bruno Ribeiro, Elisa Bertino, Ninghui Li.
Manuscript under review, 2026.

[M2] Under review **On the Security of Dataset Membership Inference Protocols against Malicious Data Owners.**
Patrick Li†, **Yuntao Du**, Neil Gong.
Manuscript under review, 2026.

[M1] Technical report **AutoVerifier: An Agentic Automated Verification Framework Using Large Language Models.**
Yuntao Du, Minh Dinh, Kaiyuan Zhang, Ninghui Li.
Winner of the 2025-2026 Radiance Technologies Innovation Bowl, arXiv:2604.02617, 2026.

Patents

U.S. Patent **Explainable Cluster-based Collaborative Filtering.**
Mingqi Wu, Jianxun Lian, **Yuntao Du**, Jing Yao, Xiting Wang, Bei Lu, Xing Xie.
U.S. Patent No. 12,524,502 B2, 2026.

Selected Media Coverage and Research Impact

AutoProfiler [P21]
ACL Findings 2026

Covered in **MIT Technology Review**:
- How LLMs could supercharge mass surveillance in the US (April 2026).
- AI chatbots are giving out people's real phone numbers (May 2026).

AutoLocate [P22]
CCS 2026

QGIS: findings incorporated into its privacy guidance.
U.S. CDC: working with the agency to revise its cartographic guidelines.

AutoVerifier [M1]
Technical report, 2026

Quantum Zeitgeist: AI System Verifies Technical Claims Without Expert Knowledge (April 2026).
Purdue CS News: coverage of the Innovation Bowl win (May 2026).

SOFT [P13]
USENIX Security 2025

Covered in **JetBrains Research**: Membership Inference Attacks and the Privacy Cost of Fine-Tuning (June 2026).

Teaching

Instructor, CS 556: Data Security and Privacy, Purdue University. Fall 2026
- *Online offering with 3 graduate students.*
- *Redesigned selected course modules, delivered lectures, and developed assignments and examinations.*
- *Extended the course with new content on AI security and privacy.*

Sole Teaching Assistant, CS 556: Data Security and Privacy, Purdue University. Spring 2025

- *Supported 18 graduate students in the CS and ECE departments.*
- *Held office hours, coordinated course projects, and graded assignments/exams.*

Mentoring

Edward Chen, high school student, Carmel High School. 2026 – Present
Publications: CCS 2026 poster paper [S1]

Patrick Li, B.S. Computer Science, Stanford University. 2025 – Present
Publications: 1 manuscript under review [M2]

Tanishq Pauskar, B.S. Computer Science, Purdue University. 2025 – Present
Mentored through the NSF CISE Research Experiences for Undergraduates (REU) program.
Publications: CCS 2026 [P22]

Meng Tong, M.S. Computer Science, University of Science and Technology of China. 2025 – Present
Publications: USENIX Security 2026 [P19]

Hillary Yang, high school student, Carmel High School. 2024 – 2025
Now an undergraduate student at Stanford University.
Publications: 1 manuscript on arXiv

Yujia Hu, M.S. Computer Science, Zhejiang University. 2022 – 2024
Publications: SIGIR 2022 [P4], VLDB 2023 [P11], ICDE 2024 [P12]

Zhihao Zeng, M.S. Computer Science, Zhejiang University. 2022 – 2023
Publications: ICDE 2023 [P8]

Xinjun Zhu, M.S. Computer Science, Zhejiang University. 2022 – 2023
Publications: SIGIR 2022 [P3], SIGIR 2023 [P9], TKDE 2023 [P6]

Academic Service

Conference Reviewing

Program Committee (PC) Member, USENIX Security Symposium	2027
PC, International Conference on Very Large Data Bases (VLDB)	2027
PC, ACM Asia Conference on Computer and Communications Security (AsiaCCS)	2027
PC, Neural Information Processing Systems (NeurIPS)	2026
Poster Committee, IEEE Symposium on Security and Privacy (S&P)	2026
PC, International Conference on Learning Representations (ICLR)	2025 – 2027
PC, International Conference on Artificial Intelligence and Statistics (AISTATS)	2025 – 2026
PC, ACM Web Conference (WWW)	2027
PC, ACM Conference on Data and Application Security and Privacy (CODASPY)	2026 – 2027
PC, International Conference on Web Search and Data Mining (WSDM)	2026
PC, ACM Conference on Information and Knowledge Management (CIKM)	2024 – 2026
PC, ACM Conference on Research and Development in Information Retrieval (SIGIR)	2023 – 2026
PC, International Joint Conference on Neural Networks (IJCNN)	2026 – 2027
PC, Table Representation Learning Workshop (TRL)	2024 – 2025
PC, AAAI Conference on Artificial Intelligence (AAAI)	2023 – 2027
PC, ACM Conference on Information Retrieval in the Asia Pacific (SIGIR-AP)	2023 – 2024

Journal Reviewing

Reviewer, ACM Computing Surveys (CSUR)
Reviewer, IEEE Transactions on Dependable and Secure Computing (TDSC)
Reviewer, ACM Transactions on Privacy and Security (TOPS)
Reviewer, International Journal on Very Large Data Bases (VLDBJ)
Reviewer, ACM Transactions on Recommender Systems (TORS)
Reviewer, IEEE Transactions on Knowledge and Data Engineering (TKDE)

Reviewer, IEEE Transactions on Big Data (TBD)

Selected Talks and Guest Lectures

NSF Center for Distributed Confidential Computing (CDCC) Workshop 2026 Oct
Privacy Leakage from a Thousand Words: Millipixel Location Recovery from Dot Maps

CS 4501, Data Privacy, *Guest lecture*, University of Virginia 2026 Apr
Membership Inference Attacks: Progress and Directions

CS 593, Trustworthy Machine Learning, *Guest lecture*, Purdue University 2026 Mar
Membership Inference Attacks: History, Recent Progress, and Future Directions

Research Trend 2025 Jun
Automated Profile Inference with Language Model Agents

Graduate Research Symposium, Purdue University 2025 Mar
Automated Profile Inference with Language Model Agents

CS 556, Data Security and Privacy, *Guest lecture*, Purdue University 2025 Mar
Differentially Private Data Synthesis

AI Time 2023 Nov
Towards Explainable Collaborative Filtering with Taste Clusters Learning

Microsoft Research Asia 2023 Jan
Explainable Clustering and Cluster-based Collaborative Filtering

Academic References

Ninghui Li
Samuel D. Conte Professor of Computer Science, Purdue University
ninghui@purdue.edu
https://www.cs.purdue.edu/homes/ninghui/

Elisa Bertino
Samuel D. Conte Distinguished Professor of Computer Science, Purdue University
bertino@purdue.edu
https://www.cs.purdue.edu/homes/bertino/

Bruno Ribeiro
Associate Professor of Computer Science, Purdue University
ribeirob@purdue.edu
https://www.cs.purdue.edu/homes/ribeirob/